

Decoding Udemy Course Performance

An Exploratory Data Analysis of the Business Finance Ecosystem

Translating raw platform data into actionable signals for developers, marketers, and business strategists.



Empowering Decision-Makers with Data

The goal is to extract valuable insights from raw platform data to guide strategy, understand user preferences, and map industry trends within the Business Finance sector.



App/Course Developers

Identify the mechanical attributes (duration, difficulty level, lecture count) that maximize student acquisition and completion rates.



Marketers

Uncover the relationship between pricing models, user satisfaction, and review generation to optimize promotional strategies.



Business Strategists

Spot emerging niche trends (e.g., algorithmic trading) and map the competitive landscape of paid versus free educational products.

Establishing the Analytical Boundaries

Distribution Mechanics

Analyze the macro-distribution of courses, specifically comparing the volume, popularity, and characteristics of Paid versus Free course models.



The Popularity Index

Identify the market leaders. Create a definitive list of the Top 10 most popular courses based entirely on raw subscriber counts.



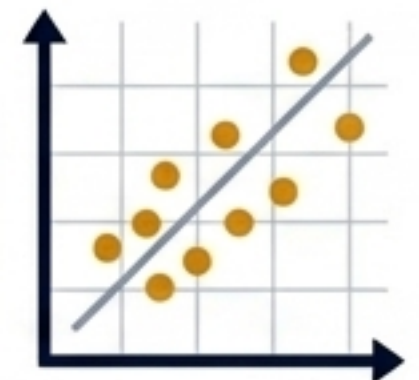
Student Engagement

Examine courses with the highest number of user reviews to understand the drivers of student engagement and post-completion satisfaction.



Price Correlation

Statistically explore the relationship between course pricing and user reviews to determine if higher costs correlate with higher perceived value.



The Analytical Pipeline

01

Data Understanding

Load raw CSV data.
Map the 12 primary features.
Filter and isolate the features required for final model building.



02

Exploratory Data Analytics (EDA)

Execute univariate and bivariate analyses to identify baseline distributions.



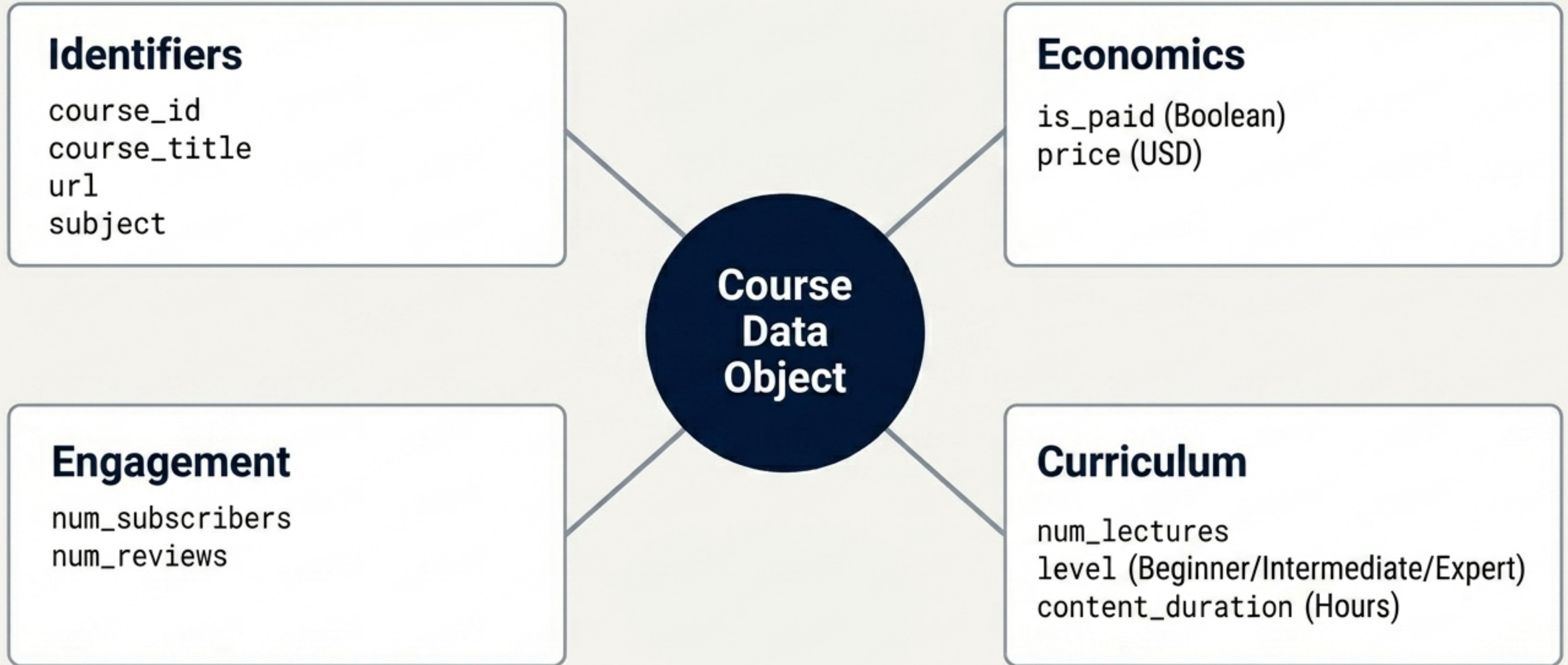
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Feature Transformation

Because Gaussian variables are used, bypass standard Z-scaling. Instead, run diagnostics to check for data skewness and apply mitigation protocols to ensure model stability.



The Anatomy of a Course Profile



The Economics of E-Learning: Paid vs. Free

Paid Courses (is_paid = True)	Free Courses (is_paid = False)
Revenue Model Generates direct capital. Prices typically range from \$20 to \$200.	Revenue Model Zero barrier to entry. Often acts as a loss-leader or top-of-funnel marketing tool.
Content Depth Generally features longer content_duration and higher num_lectures to justify the premium price point.	Content Depth Frequently shorter, acting as crash courses or introductory modules.
Audience Targets highly intent-driven learners willing to invest financially.	Audience Dominates raw subscriber volume, capturing casual learners and beginners.

The Market Leaders: Top 10 by Popularity

Ranked exclusively by num_subscribers.

1	Bitcoin or How I Learned to Stop Worrying...		65,576 Subs
2	Accounting in 60 Minutes - A Brief Intro		56,659 Subs
3	Stock Market Investing for Beginners		50,855 Subs
4	Introduction to Financial Modeling		29,167 Subs
5	The Complete Financial Analyst Course 2017		24,481 Subs
6	Forex Basics		22,344 Subs
7	Beginner to Pro in Excel: Fin Modeling...		22,257 Subs
8	Black Algo Trading: Build Your Robot		20,195 Subs
9	Financial Analysis: A Recipe for Success		19,614 Subs
10	Stock Market Foundations		19,339 Subs

Analyzing the Leaders: The Zero-Price Barrier

1	Bitcoin or How I Learned to Stop Worrying...	65,576 Subs
2	Accounting in 60 Minutes - A Brief Intro	56,659 Subs
3	Stock Market Investing for Beginners	50,855 Subs
4	Introduction to Financial Modeling	29,167 Subs
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70%
of the Top 10 are
completely free.

While paid courses dominate **revenue**, **volume acquisition** is overwhelmingly driven by **free introductory content**. Courses like Bitcoin or How I Learned... and Accounting in 60 Minutes leverage the \$0 price point to amass tens of thousands of users, functioning as massive top-of-funnel magnets.

The Engagement Champions

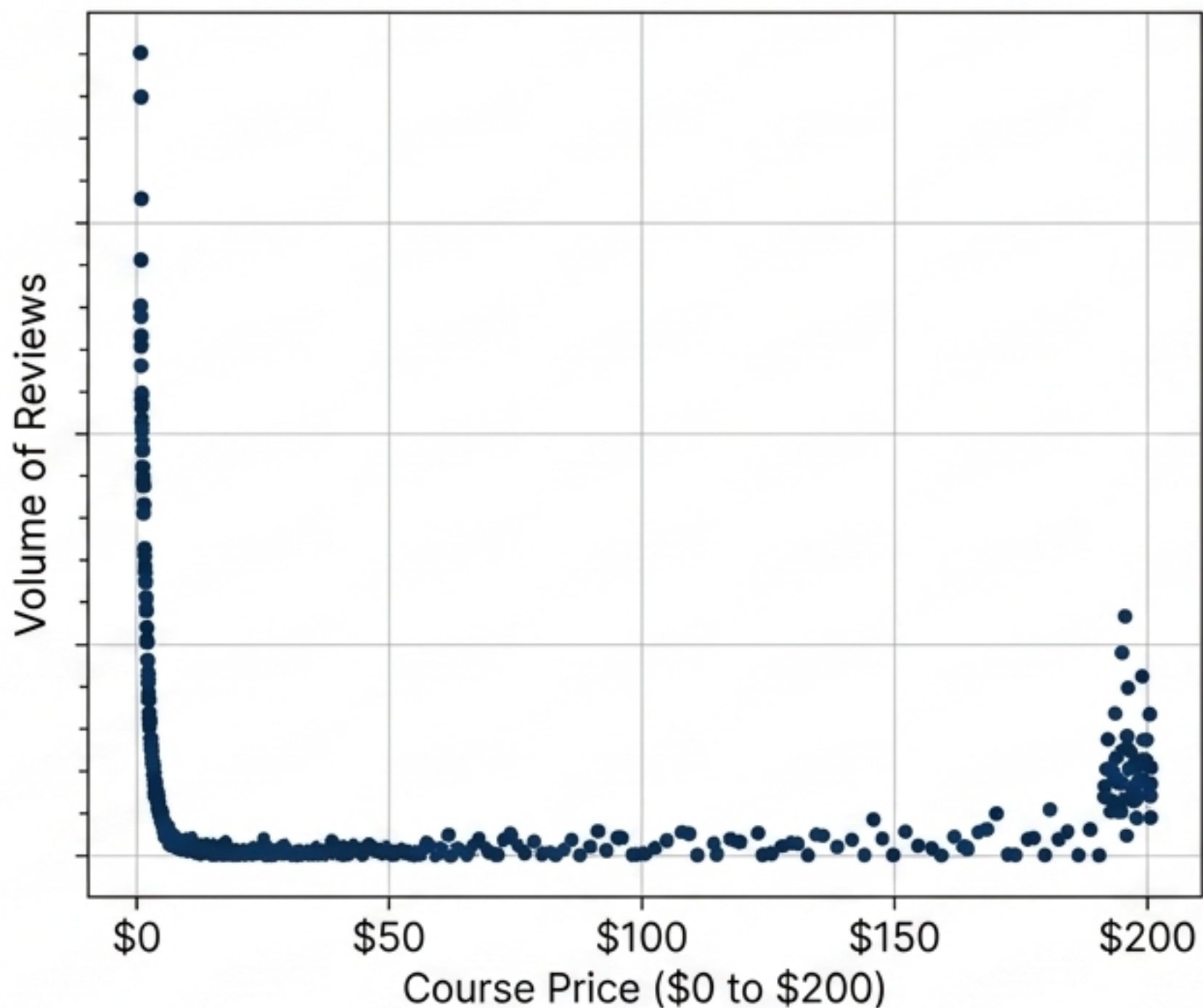
Filtering the dataset for maximum num_reviews to identify highest active student satisfaction.



High review counts do not solely belong to free courses. Premium courses (\$195) successfully **generate massive engagement** when they deliver **comprehensive, professional**-grade curricula (e.g., 138 **lectures**, 7.5 hours duration).

Hypothesis: Does Price Drive Engagement?

Takeaway: The raw EDA reveals a non-linear relationship.



The Free Anomaly

\$0 courses generate **extreme review volumes** due to massive subscriber bases, not necessarily a higher completion rate.

The Premium Squeeze

Mid-tier pricing (\$20-\$100) often struggles to generate high engagement.

The Flagship Effect

The **highest prices** (\$195-\$200) generate dense review clusters, suggesting that learners who make significant financial investments are **highly committed** to course completion and feedback.

Niche Spotlight: The Python Ecosystem

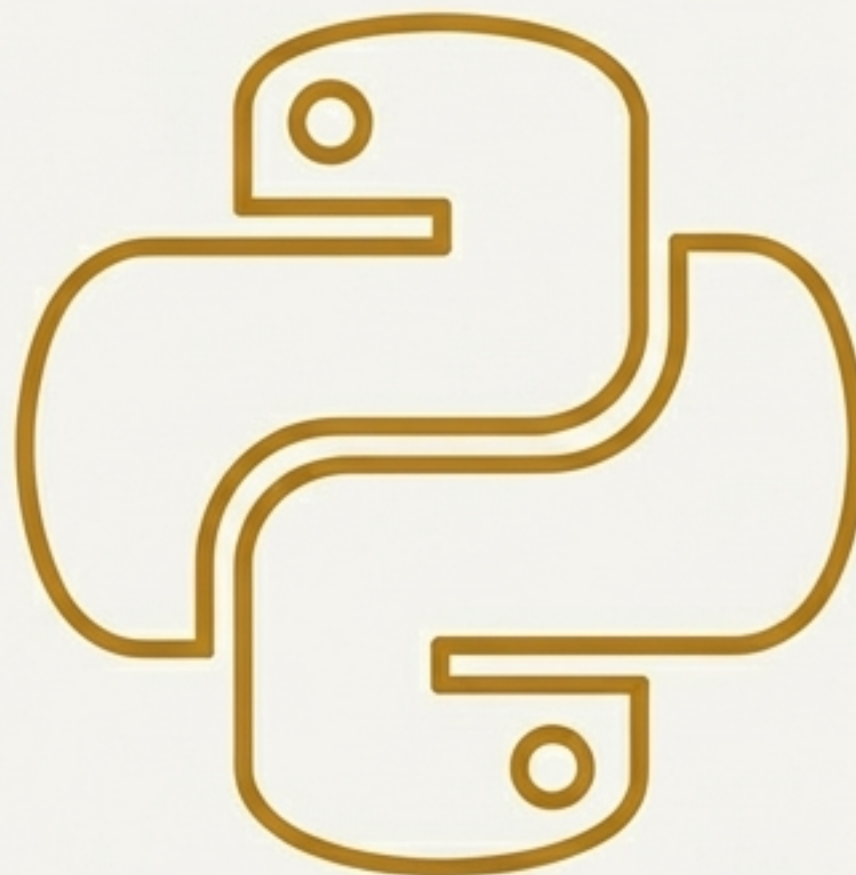
Analyzing the performance of algorithmic and data-science courses within the Business Finance sector.

Python for Finance: Investment Fundamentals

\$195

3,811
Subs

278
Reviews



Python Algo Stock Trading: Automate Your Trading!

\$95

1,165
Subs

21
Reviews

Python Algo Trading: Sentiment Trading

\$200

294
Subs

19
Reviews

Stock Technical Analysis with Python

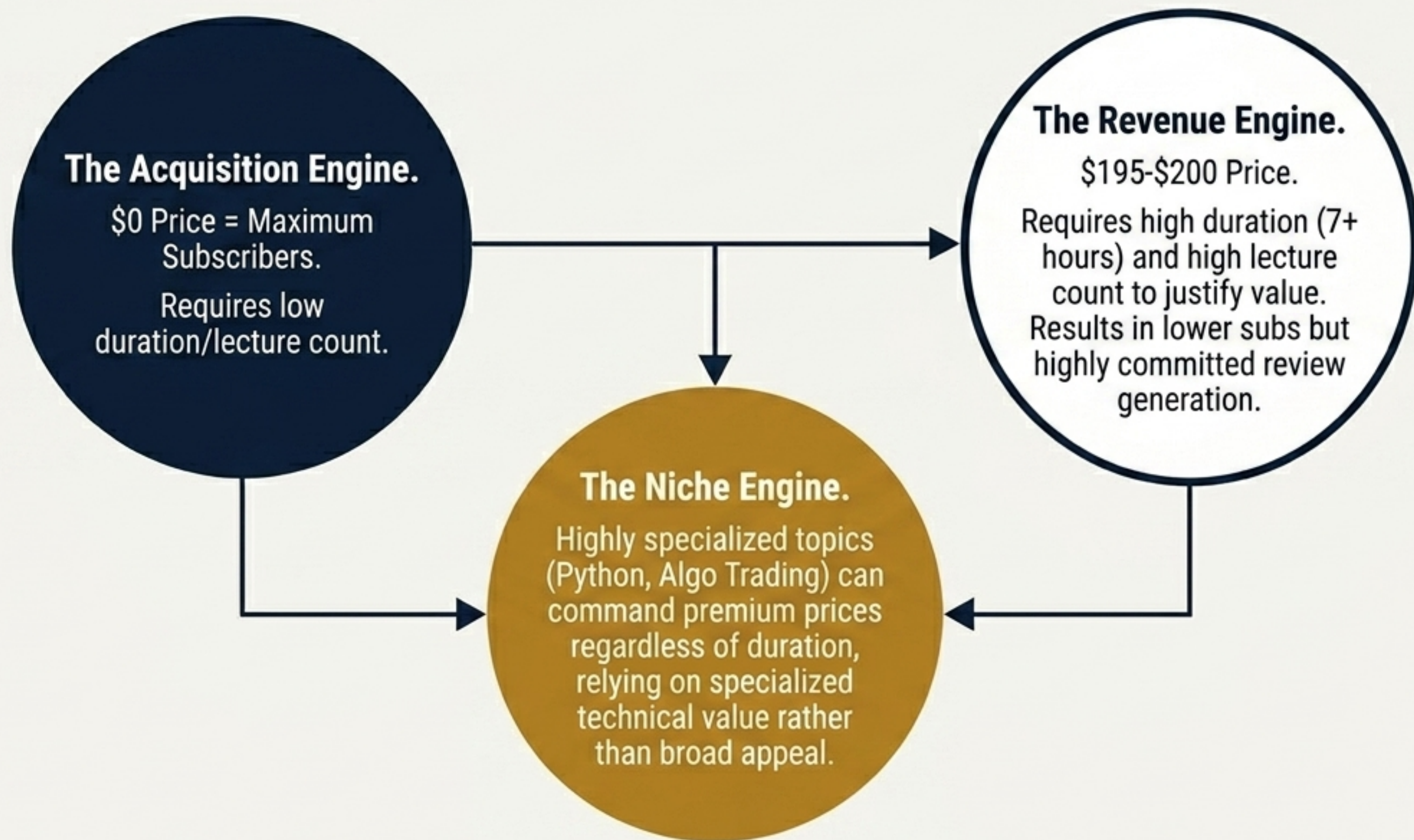
\$50

409
Subs

35
Reviews

Python courses represent a high-value, intermediate-to-expert niche. While they do not reach the massive volume of basic accounting or free investing courses, they command **premium pricing** (\$95-\$200) and attract highly targeted, serious learners.

Synthesizing the Course Ecosystem



Next Steps: Preparing for Predictive Modeling



Skewness Mitigation



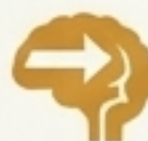
The EDA phase revealed significant skewness (e.g., a few courses hold the vast majority of subscribers and reviews). This must be normalized to prevent predictive model bias.

Feature Selection



Finalizing the isolation of Gaussian variables.

Model Building



Transitioning from historical analysis to building algorithms capable of predicting course success based on title, duration, and price.

Key Strategic Takeaways



For Marketers: Leverage Loss-Leaders

Use free, short-duration courses as high-volume lead magnets, recognizing that 70% of the top 10 market leaders utilize this strategy.



For Developers: Justify the Premium

Mid-tier pricing struggles. If charging \$150+, the curriculum must scale accordingly in lecture count and hours to drive the high high review engagement seen in top paid courses.



For Strategists: Target the Technical Niche

Courses blending traditional finance with modern data science (Python, Algo Trading) command high prices and attract a dedicated, intent-driven user base.